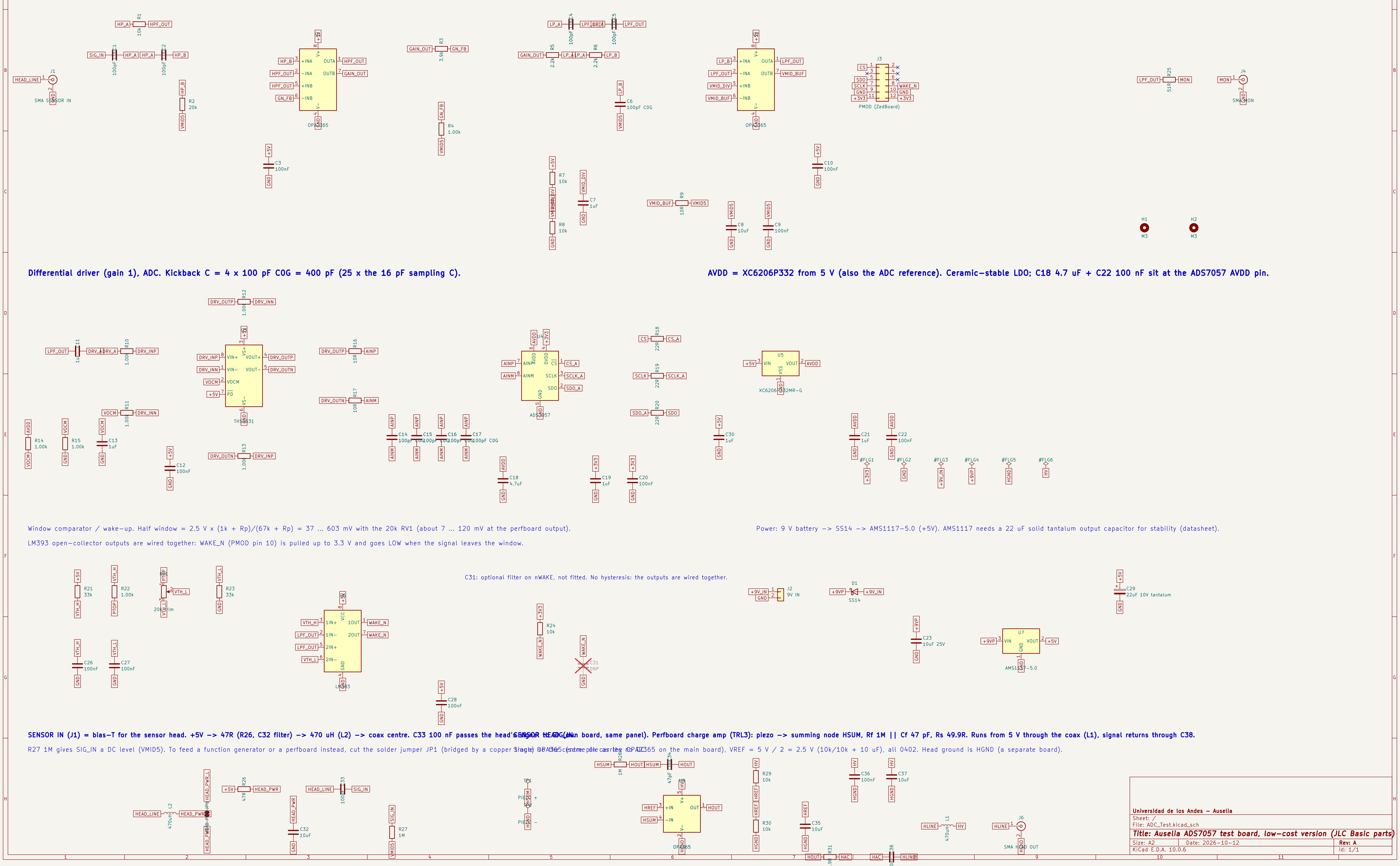
ADS7057 test board (low cost)

Sensor head (SMA, bias-T) -> 2nd-order HPF 113 kHz -> gain x4.9 -> 2nd-order LPF 512 kHz -> THS4531 -> 10R/400pF -> ADS7057 -> ZedBoard PMOD.

AFE v0.1 topology rebuilt around JLC Basic parts (100 pF C0G, E12 resistors, AMS1117, LM393). Power: 9 V -> AMS1117-5.0 -> AMS1117-3.3 (AVDD). DVDD = PMOD 3.3 V.



Input, high-pass filter (Sallen-Key, 100 pF / 10k / 20k: f0 = 113 kHz, Q = 0.71), gain 1 + 3.9k/1k = 4.9

Low-pass filter (2.2k / 2.2k, C1 = 2 x 100 pF, C2 = 100 pF: f0 = 512 kHz, Q = 0.71), VMID5

ZedBoard PMOD JA (3.3 V). Pin 6/12 = 3.3 V from the ZedBoard -> DVDD. Pin 10 = WAKE_N (active low).

MON: LPF_OUT through 51R to an SMA for the scope (same signal

Differential driver (gain 1), ADC. Kickback C = 4 x 100 pF C0G = 400 pF (25 x the 16 pF sampling C).

AVDD = XC6206P332 from 5 V (also the ADC reference). Ceramic-stable LDO; C18 4.7 uF + C22 100 nF sit at the ADS7057 AVDD pin.

Window comparator / wake-up. Half window = 2.5 V x (1k + Rp)/(67k + Rp) = 37 … 603 mV with the 20k RV1 (about 7 … 120 mV at the perfboard output).

LM393 open-collector outputs are wired together: WAKE_N (PMOD pin 10) is pulled up to 3.3 V and goes LOW when the signal leaves the window.

Power: 9 V battery -> SS14 -> AMS1117-5.0 (+5V). AMS1117 needs a 22 uF solid tantalum output capacitor for stability (datasheet).

C31: optional filter on nWAKE, not fitted. No hysteresis: the outputs are wired together.

SENSOR IN (J1) = bias-T for the sensor head. +5V -> 47R (R26, C32 filter) -> 470 uH (L2) -> coax centre. C33 100 nF passes the head's signal to SIG_IN.

SENSOR HEAD (own board, same panel). Perfboard charge amp (TRL3): piezo -> summing node HSUM, Rf 1M || Cf 47 pF, Rs 49.9R. Runs from 5 V through the coax (L1), signal returns through C38.

R27 1M gives SIG_IN a DC level (VMID5). To feed a function generator or a perfboard instead, cut the solder jumper JP1 (bridged by a copper trace) [illegible]

Single OPA365 (same die as the OPA2365 on the main board), VREF = 5 V / 2 = 2.5 V (10k/10k + 10 uF), all 0402. Head ground is HGND (a separate board).

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Title: Auselia ADS7057 test board, low-cost version (JLC Basic parts)

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